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High Thermal Stability and Low Thermal Resistance of Large Area GaN/3C-SiC/Diamond Junctions for Practical Device Processes

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Thermal management is critical in contemporary electronic systems, and integrating diamond with semiconductors offers the most promising solution to improve heat dissipation. However, developing a technique that can fully exploit the high thermal conductivity of diamond, withstand high-temperature annealing processes, and enable mass production is a significant challenge. In this study, the successful transfer of AlGaN/GaN/3C-SiC layers grown on Si to a large-size diamond substrate is demonstrated, followed by the fabrication of GaN high electron mobility transistors (HEMTs) on the diamond. Notably, no exfoliation of 3C-SiC/diamond bonding interfaces is observed even after annealing at 1100 °C, which is essential for high-quality GaN crystal growth on the diamond. The thermal boundary conductance of the 3C-SiC-diamond interface reaches ≈55 MW m^{-2} K^{-1}, which is efficient for device cooling. GaN HEMTs fabricated on the diamond substrate exhibit the highest maximum drain current and the lowest surface temperature compared to those on Si and SiC substrates. Furthermore, the device thermal resistance of GaN HEMTs on the diamond substrate is significantly reduced compared to those on SiC substrates. These results indicate that the GaN/3C-SiC on diamond technique has the potential to revolutionize the development of power and radio-frequency electronics with improved thermal management capabilities.

1. Introduction

The increasing miniaturization and integration of semiconductor devices, as well as the increasing adoption of Artificial Intelligence (AI) and 5th Generation (5G) communication technology, have led to an increase in power density and heat generation.[1–3] Therefore, effective thermal management has become crucial to ensure device performance and reliability. To this end, researchers are exploring various techniques and materials, including advanced packaging technologies,[4] high thermal conductivity materials,[5,6] and innovative cooling systems,[7] to enhance thermal management and improve device efficiency and reliability. Moreover, the development of new chip architectures and designs that prioritize heat dissipation can also contribute to effective thermal management.[8–10] The advancement of technology and the increasing demand for higher performance and efficiency have made thermal management a critical issue in the development of semiconductor devices. Effective thermal management techniques and materials are essential for improving device performance, reliability, and lifetime.

Diamond has the highest thermal conductivity among all natural materials, making it an excellent heat-spreading substrate for high-power and high-frequency semiconductor devices, and has been extensively studied for the cooling of GaN high electron mobility transistors (HEMTs).[11–14] The integration of diamonds with GaN HEMTs is still challenging and requires advanced processing techniques and materials. The GaN-on-diamond

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structure has been developed by either removing Si from the GaN-on-Si substrate and depositing diamond on the exposed GaN surface using a dielectric material transition layer such as SiNx or AlN,[13,15,16] or removing Si from the GaN-on-Si substrate and bonding it to a diamond substrate using an adhesion layer such as metal, Si, and SiC.[17–19] Although a 4-inch GaN-on-diamond wafer has been achieved in the former case,[20] the transition layer contributed to a large thermal resistance due to its lower thermal conductivity, and the low crystal quality of the diamond deposited on the transition layer resulted in low thermal conductivity,[21,22] which prevents the desired thermal management from being achieved. In the latter case, the "device first" concept was used,[23,24] which involves fabricating the GaN devices on a Si substrate and then bonding the GaN to the diamond substrate at high temperature. This technique can introduce unevenness and/or ununiform strain during GaN device fabrication processes,[25] which have a significant negative impact on the performance and reliability of the devices. Therefore, it is unsuitable for large-diameter wafer applications with critical uniformity and consistency.

Transferring GaN layers to a diamond substrate before device fabrication can mitigate the challenges associated with the "device first" concept. This approach, known as the "bonding first" concept, enables the fabrication of GaN devices on a flat and uniform surface, leading to better performance and reliability. Moreover, this concept is suitable for large-diameter wafers, which is critical for the mass production of GaN-based devices. However, the primary challenges in this process revolve around the high-temperature annealing and the imperative to minimize thermal boundary resistance at the interface. To achieve ohmic contacts, the sample must undergo a high-temperature annealing process of ≈800 °C.[26,27] Such high temperatures can potentially compromise the thermal stability of the bonding interface and introduce thermal stress, cracking, and other defects due to the differences in crystal structures and thermal expansion coefficients between GaN and diamond. It is also worth noting that the epitaxial growth of GaN typically requires high temperatures around 1100 °C,[28,29] which can further complicate the bonding interface. M. Trejo, et al. reported the fabrication of GaN HEMTs on diamond through the bonding-first approach.[30] However, the improvement in thermal management observed in the fabricated GaN HEMTs on the diamond was significantly constrained by the relatively lower thermal conductivity of the proprietary adhesion layer.

3C-SiC, being a cubic polytype of silicon carbide, exhibits a thermal expansion coefficient that lies between that of GaN and diamond.[31] The lattice constant of 3C-SiC (0.436 nm) closely matches that of GaN (0.45 nm). This alignment allows for the fabrication of large-diameter substrates, simplifying the process of growing high-quality GaN layers using basic buffers.[32] It is worth noting that the 3C-SiC/diamond interface displayed noteworthy thermal stability even after undergoing annealing at 1000 °C.[33] This feature positions 3C-SiC as an ideal solution for mitigating the challenges related to lattice mismatch during bonding processes, ultimately facilitating seamless integration of GaN and diamond materials. Furthermore, corresponding 3C-SiC thin films have been reported to possess record-high thermal conductivity[34]. The computation of interfacial thermal resistance by the phonon diffuse mismatch model indicated that the thermal boundary resistance for the 3C-SiC/diamond interface is comparatively lower when compared to other diamond heterointerfaces.[35] Therefore, utilizing 3C-SiC as an intermediate layer for bonding GaN and diamond presents a promising solution to address the challenges associated with the "device first" concept.[36] This technique offers several advantages, including enhanced thermal management, improved crystal growth of GaN, and reduced thermal stress and defects at the bonding interface. However, if further research and development are conducted, this technique has the potential to enable the mass production of GaN-based devices on diamond substrates for a wide range of applications. By exploring and refining this technology, it is possible to overcome existing challenges and achieve significant advancements in the field. The successful implementation of GaN-based devices on diamond substrates could have far-reaching implications, offering enhanced performance, improved thermal management, and increased reliability for various electronic and optoelectronic applications.

In this study, we report the implementation of GaN HEMTs on diamond substrates using the "bonding first" approach. The experimental process involved the transfer of AlGaN/GaN/3C-SiC layers grown on Si substrates to a diamond substrate. Subsequently, GaN HEMTs were fabricated using a conventional process. Rapid thermal annealing experiments were conducted at different temperatures within a high vacuum chamber to evaluate the thermal stability of the AlGaN/GaN/3C-SiC/diamond bonded samples. The purpose of these experiments was to investigate the effects of thermal treatment on the structural integrity and performance of the bonded GaN HEMTs on diamond substrates. The residual stress of GaN and 3C-SiC in the AlGaN/GaN/3C-SiC/diamond bonded samples after annealing at different temperatures was characterized by the micro-Raman spectroscopy technique. The bonding interface between 3C-SiC and diamond was examined using transmission electron microscopy (TEM) and energy-dispersive X-ray spectroscopy (EDS). Furthermore, electron energy-loss spectroscopy (EELS) was employed to systematically investigate the chemical bonding states of the interface. Furthermore, the thermal conductivity (κ) of the 3C-SiC layer and thermal boundary conductance (TBC) were investigated. Finally, we compared the electrical properties and surface temperature during the operation of GaN HEMTs on the diamond with those of GaN HEMTs fabricated on Si and SiC substrates.

2. Experimental Section

The heterostructure consisting of multiple layers grown on a Si (111) substrate using a metal-organic chemical vapor deposition (MOCVD) was used in this study. The layers included a 23 nm-thick $Al_{0.26}Ga_{0.74}N$ barrier layer, a 2 µm-thick GaN layer, a 1 µm-thick AlGaN/AlN buffer layer, and a 1 µm-thick 3C-SiC layer. The heterostructure was temporarily bonded to a carrier substrate, and then the Si substrate was removed using the fluoric acid etching process. The exposed 3C-SiC surface was polished using a chemical mechanical polishing technique to get a flat surface, as depicted in Figure S1 (Supporting Information). Subsequently, it was cleaned with acetone and isopropyl alcohol, followed by drying under N_2 flow. After cleaning, the 3C-SiC surface was bonded to diamond substrates, which were

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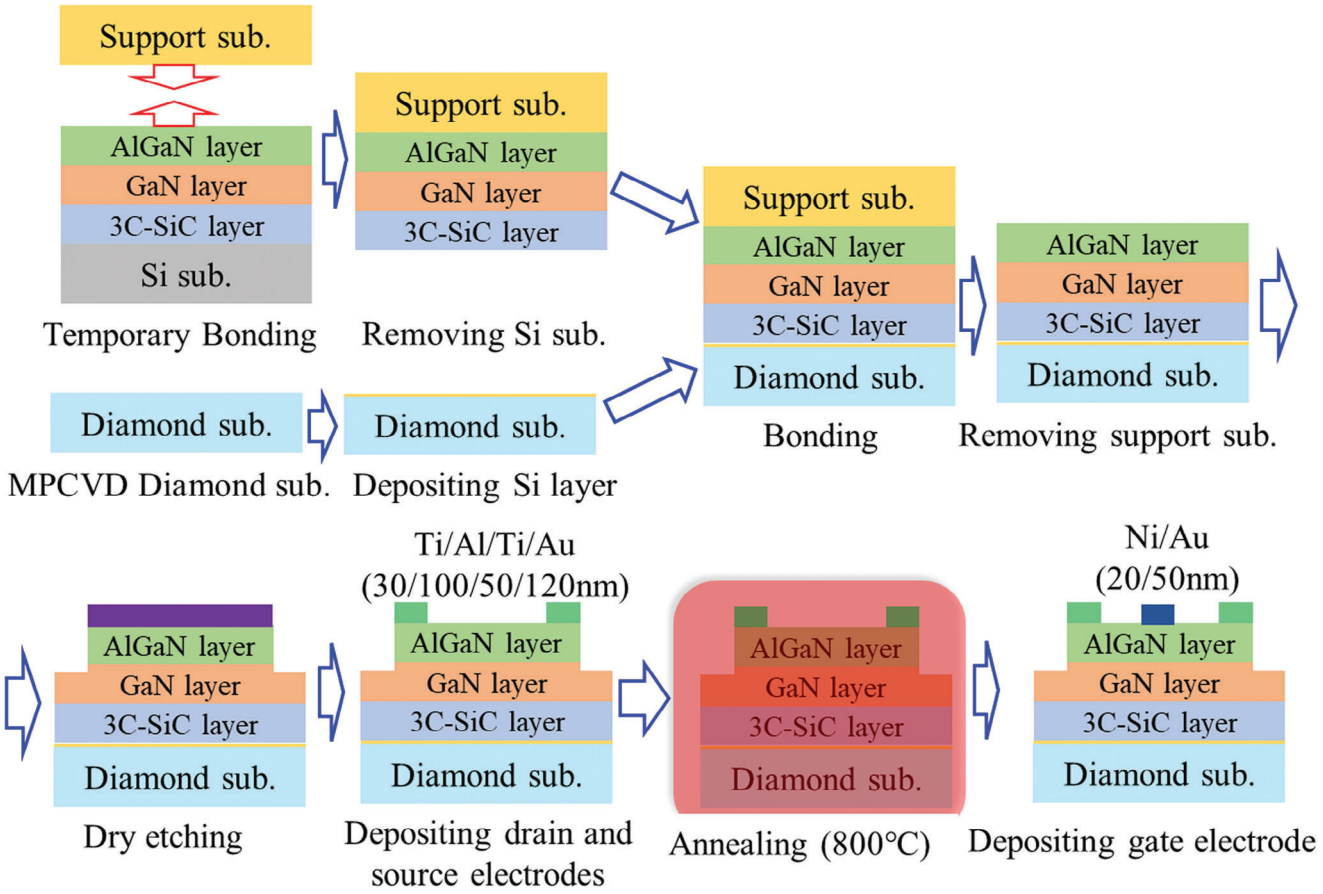


Figure 1. The transfer process of the AlGaN/GaN/3C-SiC layers and the fabrication process of GaN HEMTs using the bonding-first concept.

synthesized via microwave plasma chemical vapor deposition (MPCVD) and came in two different sizes: 10 mm × 10 mm × 0.45 mm and 20 mm × 20 mm × 0.45 mm. Before bonding, the MPCVD diamond substrates were polished using a mechanical polishing technique and then cleaned with the sulfuric acid-hydrogen peroxide mixture. The atomic force microscopy (AFM) revealed that the diamond surface exhibited an average roughness (Ra) of 0.37 nm, as illustrated in Figure S1 (Supporting Information). The bonding process was carried out at room temperature using the surface-activated bonding (SAB) method. Before bonding, an a-Si (amorphous silicon) layer with a thickness of ≈15 nm was deposited on the diamond surface by RF sputtering. After bonding, the carrier substrate was removed, and the AlGaN/GaN/3C-SiC layers were transferred onto the diamond substrate.

After the transfer process, the sample with a size of 10 mm × 10 mm × 0.45 mm was used for the interfacial characterization and thermal stability testing, while the sample with a size of 20 mm × 20 mm × 0.45 mm was used for fabricating GaN HEMTs on diamond using a conventional process. The fabrication process of GaN HEMTs on diamond includes mesa isolation by Ar ion milling dry etching technique, formation of source and drain ohmic contacts by evaporating Ti/Al/Ti/Au multilayers and annealing at 800 °C for 60 s in N_2 ambient, and formation of gate Schottky contacts by evaporating Ni/Au layers. The same process was used to fabricate GaN HEMTs on a 4H-SiC substrate, while GaN HEMTs on a Si substrate were fabricated using a conventional process. The transfer process of the AlGaN/GaN/3C-SiC layers and the fabrication process of GaN HEMTs using the bonding-first concept are shown in **Figure 1**. More detailed processes of diamond bonding can be found in the previous work.[37]

To evaluate the thermal stability of the bonding interface, rapid thermal annealing was carried out using a GVL298 rapid thermal annealing furnace from Thermo Riko Co., Ltd. The annealing process was performed at various temperatures, namely 600, 800, 950, and 1100 °C, for 10 min. During the annealing process, the temperature was ramped up at a rate of 60 °C sec^{-1}, and once the desired temperature was reached, it was maintained for 10 min. Subsequently, the cooling process took place at a rate of 300 °C sec^{-1}. To maintain a controlled testing environment, the annealing process was conducted under vacuum conditions with a pressure of 5.0×10^{-6} Pa. This ensured that the annealing process occurred in a controlled atmosphere without the presence of contaminants. After annealing at 1100 °C for 10 min, the bonding state at the interface was assessed using scanning acoustic microscopy (SAM). The SAM measurements were performed using a Hitachi FineSAT FS300III system operating at a frequency of 200 MHz with a pitch of 10 µm. SAM allowed for the visualization and characterization of the bonding interface and provided insights into the integrity and quality of the bond.

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The stress states of GaN and 3C-SiC in the AlGaN/GaN/3C-SiC on the diamond substrates were analyzed using micro-Raman spectroscopy at room temperature. Raman measurements were carried out in backscattering geometry, utilizing a 488 nm Ar laser as the excitation source. A high numerical aperture (NA) 0.55 45× objective lens was utilized to focus the laser and collect the Raman scattered light, resulting in a laser spot size of 1 µm on the sample surface. The peak positions of GaN and 3C-SiC were determined by applying Lorentz fitting of the Raman peaks. This allowed for the precise determination of the peak positions, which provided information about the stress levels in the materials. The Raman spectroscopy measurements were performed not only on the AlGaN/GaN/3C-SiC on diamond substrates but also on the AlGaN/GaN/3C-SiC on Si samples. Additionally, stress-free 3C-SiC and GaN substrates were included in the measurements to serve as reference samples for comparison.

The structure and chemical composition of the 3C-SiC/diamond bonding interface after annealing at 1100 °C for 10 min were investigated using scanning transmission electron microscopy (STEM) with a JEOL JEM-2200FS analytical microscope at an acceleration voltage of 200 kV. The TEM samples were prepared using a focused ion beam (FIB) technique with a Helios NanoLab 600i Dualbeam system from Thermo Fisher Scientific, at room temperature. To protect the samples during the FIB preparation, a protective layer of Pt was deposited. The FIB milling process was carried out using an accelerated voltage of 30 kV, and the final etching was performed with an accelerated voltage of 2 kV. This technique allowed for the precise fabrication of TEM samples with high spatial resolution. The STEM analysis was performed at an acceleration voltage of 200 kV, which provided detailed information on the structure and chemical composition of the 3C-SiC/diamond bonding interface. Energy-dispersive X-ray spectroscopy (EDS) was used to analyze the chemical composition of the interface, providing elemental information on the distribution of different elements within the interface region. EELS was used to investigate the bonding states of carbon atoms near the interface after annealing at 1100 °C. Carbon K-shell edge spectra were captured in the energy range of 270–370 eV, using an acceleration voltage of 200 kV. Additionally, line scans of EELS spectra were performed across the interface, with a step size of 0.31 nm.

The κ and TBC of GaN HEMTs on the diamond were determined using the time-domain thermoreflectance (TDTR) technique. Prior to TDTR measurements, a ≈90-nanometer-thick layer of aluminum is deposited on the sample surface as the TDTR transducer. TDTR was an ultra-fast laser-based pump-probe technique capable of assessing the thermal properties of both bulk and nanostructured materials.[34] The pump beam periodically heated the sample surface while the probe beam detected the temperature variation of the sample surface via thermoreflectance. The signal picked up by a photodetector and a Lock-in amplifier was fitted with an analytical heat transfer solution of the sample structure to infer unknown parameters.[38] The picosecond ultrasonics was used to determine the Al and 3C-SiC thicknesses.

The temperature measurements of the gap between the source and gate, and the gap between the gate and drain in GaN HEMTs fabricated on the Si, SiC, and diamond substrates were performed using the micro-photoluminescence (µ-PL) technique. In the experiment, a 325 nm He-Cd laser with a power of 0.1 mW was utilized as an excitation light source. The laser beam was focused on a sport with a diameter of 1 µm using a 45×ultraviolet objective. PL spectra were then captured using a Si charge-coupled device camera. To establish the calibration of the PL spectra to the ambient temperature, the peak energies of the PL spectra were measured for GaN HEMTs under no bias applied at various temperatures. These data points were fitted using the Varshini equation,[39,40] which is a mathematical equation commonly used for calibration purposes. Please refer to Figure S2 (Supporting Information) for the details of the calibration process. It was important to note that the temperature error associated with the measurements was ±5 °C, as indicated in Figure S3 (Supporting Information). To maintain a constant ambient temperature during the temperature measurements, the GaN HEMTs on the Si, SiC, and diamond substrates were mounted on a heat sink system made of copper. The heat sink system was specifically designed to provide effective heat dissipation. To ensure a fixed temperature of 300 K, a water-cooling system was employed to actively cool the heat sink. This water-cooling system helped maintain a stable and controlled temperature environment throughout the experiment. The GaN HEMTs on the Si, SiC, and diamond substrates were biased at the same source-drain voltage as the temperature was measured. The effect of different laser powers on the surface temperature of GaN HEMTs on the diamond substrate was investigated, as shown in Figure S4 (Supporting Information). No laser-induced temperature rises were observed. By utilizing this setup, the ambient temperature surrounding the GaN HEMTs was kept at a constant 300 K, enabling accurate and reliable temperature measurements of the devices.

3. Results and Discussion

The optical microscopy images in **Figure** 2a–c show various stages of the bonded sample, including the carrier substrate/AlGaN/GaN/3C-SiC/diamond bonded sample, the sample after removing the carrier substrate, and the fabricated GaN HEMTs on the diamond. These images indicate that the AlGaN/GaN/3C-SiC layers were successfully transferred to the diamond substrate, and no voids were observed in the sample. Furthermore, Figure 2d,e shows cross-sectional SEM and TEM images of the GaN HEMTs fabricated on the diamond. The images indicate that the fabrication processes, including high-temperature annealing as high as 800 °C, were carried out without exfoliation or peeling of the AlGaN/GaN/3C-SiC layers bonded to the diamond substrate. These observations suggest that the bonding interface maintained its integrity during the fabrication processes, indicating a strong and thermally stable bonding between the layers and the diamond substrate. These images provide visual evidence of the successful transfer of the AlGaN/GaN/3C-SiC layers with a large area to the diamond substrate and the fabrication of GaN HEMTs on the diamond without any structural degradation or exfoliation at the bonding interface, even during high-temperature annealing.

The optical microscopy images in **Figure** 3a–d show the AlGaN/GaN/3C-SiC/diamond bonded sample before and after annealing at various temperatures, namely 600, 800, 950, and 1100 °C. Despite some exfoliation of the layers, ≈95% of the

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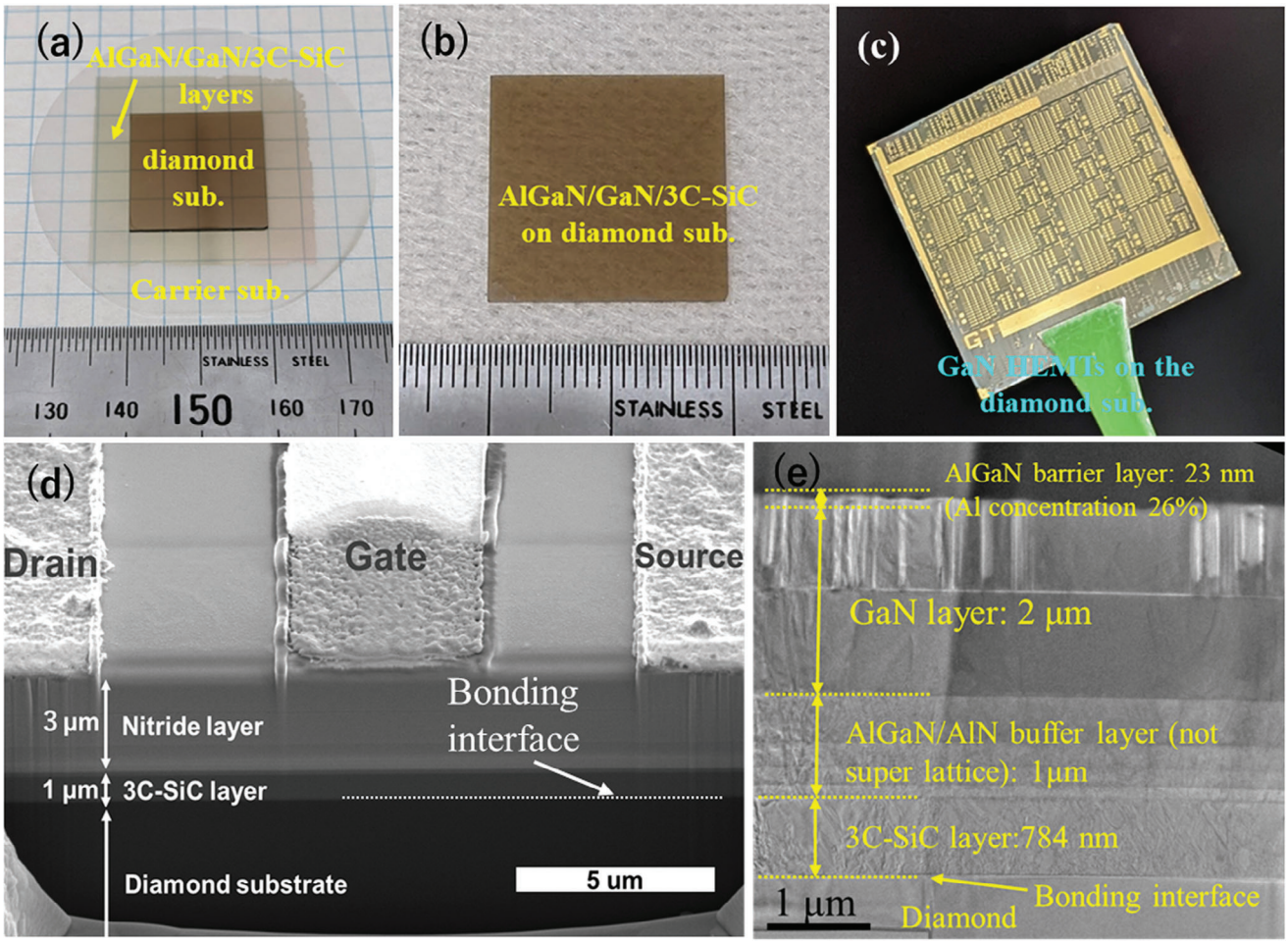


Figure 2. Optical microscopy images of a) the carrier substrate/AlGaN/GaN/3C-SiC/diamond sample, b) the sample after removing the carrier substrate, c) the fabricated GaN HEMTs on the diamond substrate, and cross-sectional d) SEM and e) TEM images of GaN HEMTs fabricated on the diamond substrate.

surface area of the layers has been successfully transferred to the diamond substrate. After annealing at 600 °C, there is no exfoliation observed in the layers bonded to the diamond, except for an area in the upper right corner where bonding is not well-formed. This suggests that the bonding is generally successful, but there are localized regions with incomplete bonding. Furthermore, after annealing at temperatures >800 °C, there are no cracks observed in the layers and no reduction in the bonded area. This indicates that the bonded layers remain intact even at high temperatures, and the bonding is stable without any significant degradation. Figure 3e in the SAM image shows that the layers are well-bonded to the diamond even after annealing at a high temperature of 1100 °C. This suggests that the bonding interface has good thermal stability and remains intact even at very high temperatures, as evidenced by the absence of cracks or exfoliation in the layers. Notably, the color of the AlGaN/GaN/3C-SiC/diamond bonded sample becomes transparent as the annealing temperature increases. This color change could be due to changes in the optical properties of the materials after annealing, which could have implications for the performance of the bonded sample in optoelectronic applications or other optical-based studies.

Figure 4a,b shows the Raman peak shift and the residual stress of GaN E_2 and 3C-SiC transverse optical (TO) phonon modes in different samples: AlGaN/GaN/3C-SiC layers grown on Si, as-bonded AlGaN/GaN/3C-SiC/diamond sample, and AlGaN/GaN/3C-SiC/diamond samples with annealing at 600, 800, 950, and 1100 °C. The residual stress values in the AlGaN/GaN/3C-SiC layers grown on Si and the AlGaN/GaN/3C-SiC layers bonded to the diamond are calculated relatively to stress-free GaN and 3C-SiC substrates, respectively. The GaN and 3C-SiC phonon frequencies are observed to shift linearly with residual stress, with coefficients of 2.7 and 3.27 cm^{-1} GPa^{-1},[41] respectively. This means that for every 1 GPa increase in residual stress, the GaN phonon frequency shifts by 2.7 cm^{-1}, and the 3C-SiC phonon frequency shifts by 3.27 cm^{-1}. The stress-free GaN substrate used in the study is a c-plane (0001) GaN single-crystal substrate grown by the hydride vapor phase epitaxy (HVPE) method. The Raman peak position of the GaN E_2 mode on this stress-free GaN substrate is observed at 568.3 cm^{-1}. It is noted that the Raman peak position of GaN grown by the HVPE method is the closest to the theoretically calculated Raman peak position.[37] On the other hand, the stress-free 3C-SiC used is a

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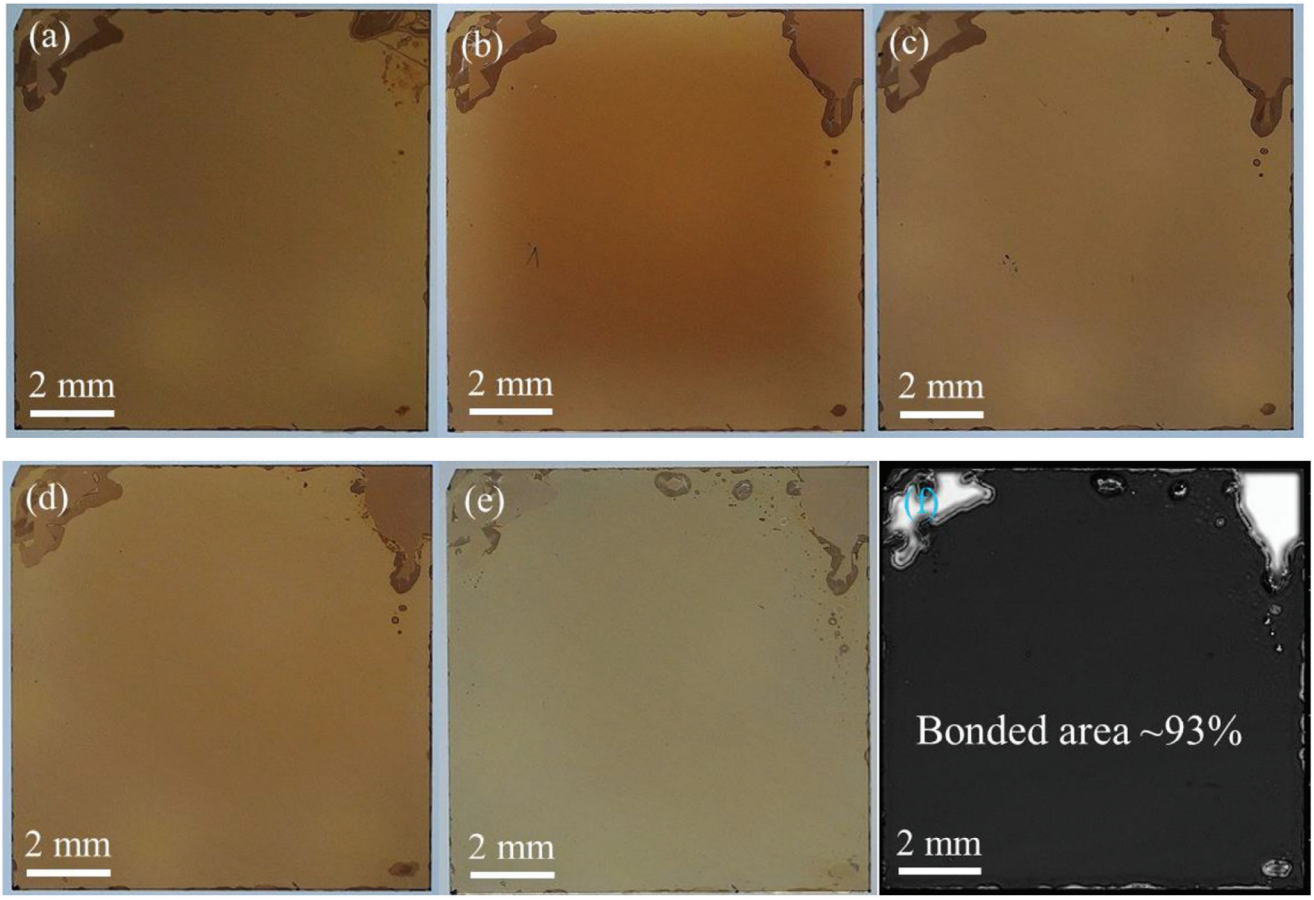


Figure 3. Optical microscopy images of the AlGaN/GaN/3C-SiC/diamond bonded sample a) before and after annealing at b) 600 °C, c) 800 °C, d) 950 °C, and e) 1100 °C, and f) SAM image of the bonded sample after annealing at 1100 °C.

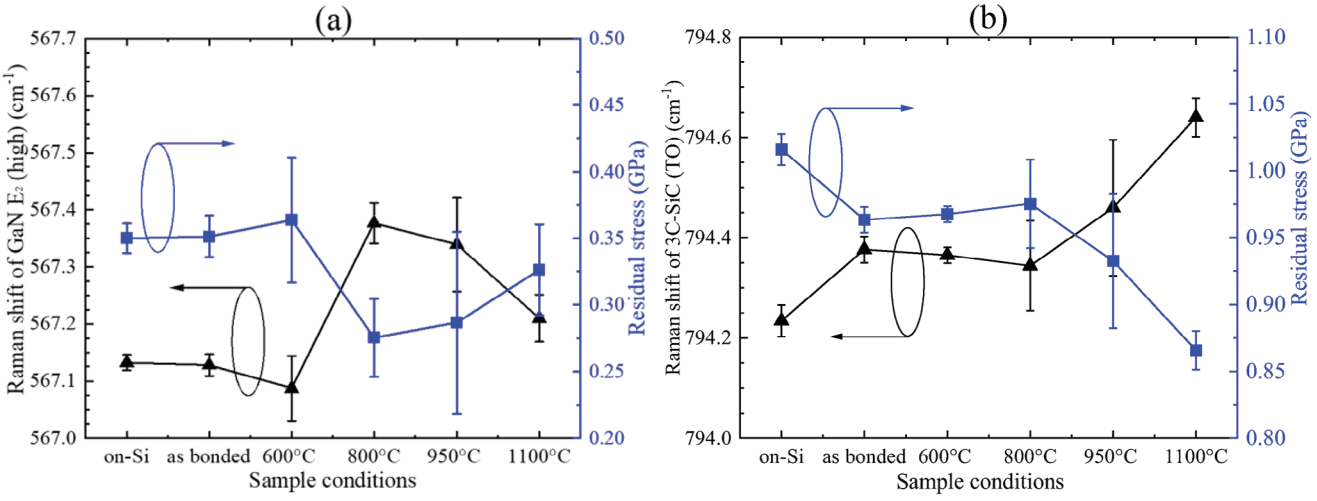


Figure 4. Raman shifts and residual stress of a) GaN and b) 3C-SiC in the AlGaN/GaN/3C-SiC layers grown on Si, as well as in the as-bonded AlGaN/GaN/3C-SiC/diamond sample, and the AlGaN/GaN/3C-SiC/diamond samples with annealing at 600, 800, 950, and 1100 °C.

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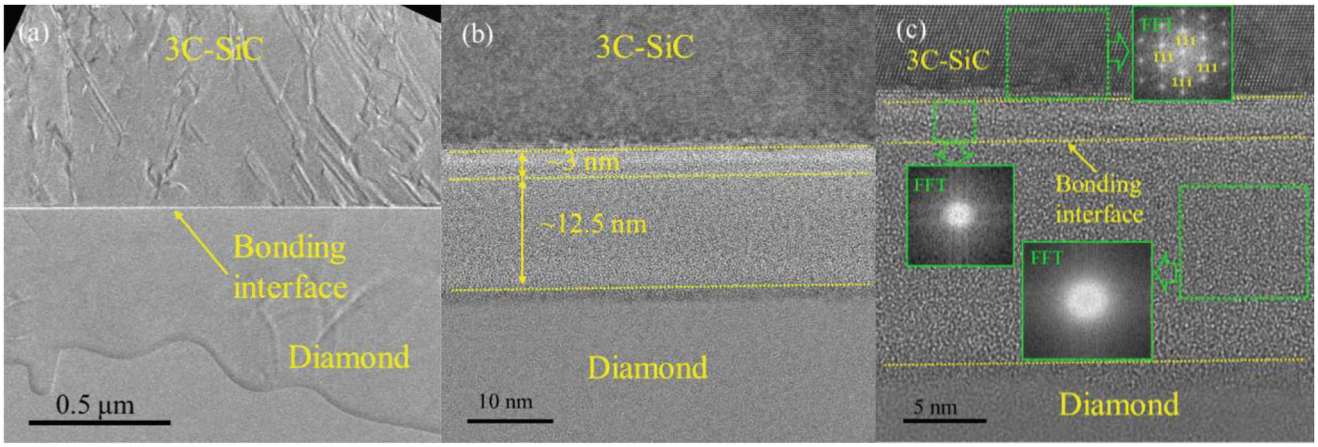


Figure 5. A cross-sectional view of the 3C-SiC/diamond interface before annealing, with low-magnification TEM (a) and HRTEM (b,c) images. FFT images are overlayed in Figure 5c.

(111) single-crystal substrate grown by chemical vapor deposition (CVD).[42,43] The Raman peak position of the stress-free 3C-SiC is observed at 797.0 cm^{-1}.

The Raman peaks of GaN and 3C-SiC in the bonded samples have shifted to lower wave numbers compared to those of the stress-free GaN and 3C-SiC substrates, indicating that the bonded samples are under tensile stress. Furthermore, annealing of the bonded samples at temperatures >800 °C resulted in a significantly smaller Raman peak shift of GaN compared to the as-bonded sample. This suggests that the annealing process may relieve some of the tensile stress in the bonded samples, resulting in a smaller peak shift toward lower wave numbers. Additionally, the Raman peak shift of 3C-SiC in the as-bonded AlGaN/GaN/3C-SiC/diamond sample is smaller compared to that of the AlGaN/GaN/3C-SiC layers grown on Si, indicating that the bonded samples have lower tensile stress in 3C-SiC compared to the layers grown on Si. Moreover, the Raman peak shift of 3C-SiC in the bonded samples tends to shift toward that of the stress-free 3C-SiC substrate with increasing annealing temperature, indicating that the annealing process can further reduce the tensile stress in 3C-SiC.

In **Figure 5a**, a cross-sectional low-magnification TEM image shows the interface between 3C-SiC and diamond before annealing. In Figure 5b,c, high-resolution TEM images provide a clear view of the interface. According to the measurements, two intermediate layers were observed at the interface. The intermediate layer adjacent to the 3C-SiC has a measured thickness of ≈3 nm, while the intermediate layer adjacent to the diamond substrate has a thickness of ≈12.5 nm. Furthermore, the application of a fast Fourier transform (FFT) to the intermediate layers confirmed that they were amorphous layers, indicating a lack of long-range order in their atomic arrangement. The amorphous layer present adjacent to the 3C-SiC was formed because of Ar fast atom beam irradiation during the bonding process. It is worth noting that the thickness of the amorphous Si layer is thinner than that of the deposited Si amorphous layer on the diamond before bonding. This reduction in thickness is attributed to the etching of the Si amorphous layer by the Ar beam irradiation during the bonding process. Importantly, there were no indications of peeling, cracks, or voids at the interface, which suggests the successful bonding between 3C-SiC and diamond. The presence of similar amorphous intermediate layers induced by the Ar fast atom beam irradiation has been reported in other interface systems, such as Si/4H-SiC,[44] Si/Si,[45] GaAs/SiC,[46] and Si/Ga_2O_3[47] fabricated using the SAB method.

Figure 6a–c shows the cross-sectional low-magnification (TEM) and high-resolution TEM (HREM) images of the 3C-SiC/diamond interface after annealing at 1100 °C. The images illustrate the presence of an intermediate layer at the interface, which was measured to have a thickness of ≈7.4 nm. Within the intermediate layer, lattice fringes corresponding to the 3C-SiC {111} planes were observed. It is interesting to note that the lattice in the layer is slightly distorted in comparison with the 3C-SiC adjacent to the interface; the spacing of the {111} planes parallel to the bonding interface is somewhat wider, while the other {111} planes are nearly normal to the interface is slightly narrower, presumably due to the Poisson effect. Figure 6c indicates that the lattice strain along [111] in the intermediate layer can be estimated to be (0.2531-0.2527)/0.2527 = +0.0016. Considering the Poisson's ratio ν of 0.168 for 3C-SiC,[48] the strain normal to [111] would be ≈−0.01. The estimated value is the same order of the average lattice mismatch between 3C-SiC {111} and diamond {001}, evaluated to be −0.017 in the Appendix. This suggests that the amorphous intermediate layer would be crystallized during 1100 °C annealing via solid-phase epitaxy at the diamond/amorphous interface, as well as at the 3C-SiC/amorphous interface, forming a pseudomorph 3C-SiC layer with lattice strain. For the heteroepitaxial growth of 3C-SiC {111} on diamond {001}, the critical thickness of 3C-SiC for generation of misfit dislocations can be estimated to be ≈7 nm (Figure S5, Supporting Information), and the estimated value is comparable to the thickness of the intermediate layer after 1100 °C annealing. This result suggests that a pseudomorph structure of 3C-SiC would be formed to relax the lattice mismatch between diamond and SiC. This observation strongly suggests that the intermediate layer is composed of crystalline 3C-SiC, which is a positive outcome for the bonding process. Additionally, the application of FFT to the intermediate layer further confirmed its composition as 3C-SiC. Overall, these findings indicate a successful bonding of 3C-SiC and diamond at the interface after annealing,

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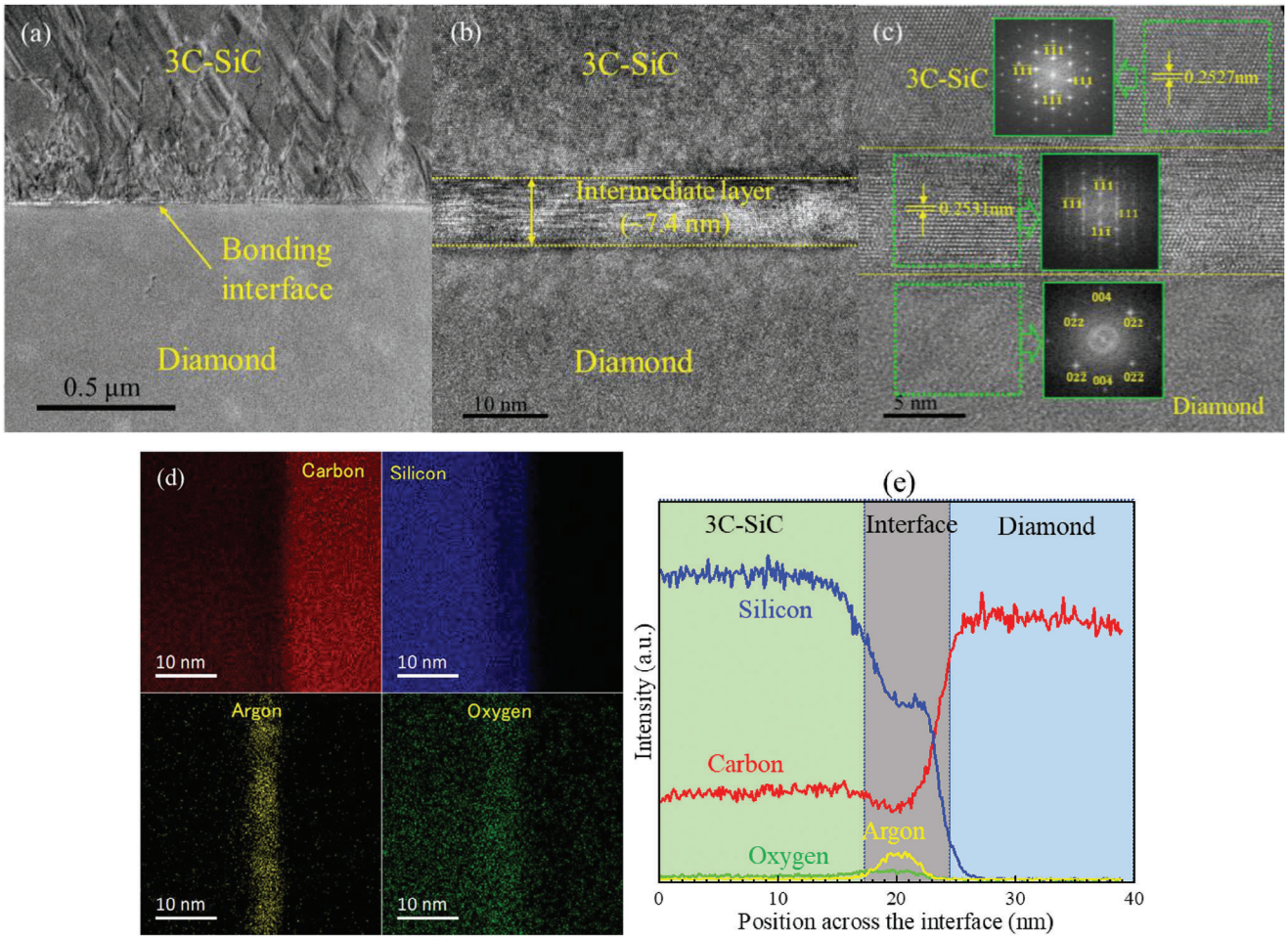


Figure 6. A cross-sectional view of the 3C-SiC/diamond bonded interface after annealing at 1100 °C, with low-magnification TEM (a) and HRTEM (b,c) images, EDS mappings of the interface (d), and X-ray intensity profiles for silicon (blue), carbon (red), oxygen (green), and argon (yellow) atoms across the interface (e). FFT images are overlayed in Figure 6c.

with the presence of a well-formed intermediate layer. This information contributes to the understanding of the structural properties and quality of the bonded interface between 3C-SiC and diamond.

EDS mappings of the interface are presented in Figure 6d, where it is evident that an argon (Ar)-rich layer was formed at the interface due to contaminants originating from the vacuum chamber during the Si film deposition processes. Ar sputtering gas was used in these processes. Figure 6e shows the X-ray intensity profiles for the silicon, carbon, oxygen, and argon atoms across the interface. Although oxygen atoms were detected at the same level as the background signal, intensity gradients were observed for silicon atoms in the intermediate layer, exhibiting a ladder-like distribution. Furthermore, the intensity of carbon atoms decreased, resulting in a concave shape at the interface. Moreover, the intensity ratio of silicon to carbon atoms at the interface is lower compared to SiC, which is located farther away from the interface. This observation indicates a deficiency of silicon atoms at the interface, preventing the formation of a perfect SiC crystal structure.

A cross-sectional HRTEM image of the 3C-SiC/diamond bonded interface after annealing at 1100 °C, an EELS line scan across the interface at the C-K edge, EELS spectra obtained at a specific distance indicated in (a), intensity profiles of the σ*and π*peaks derived from the EELS spectra as a function of the scanned distance are shown in **Figure** 7a–d, respectively. In the EELS spectra, two characteristic peaks corresponding to carbon materials were observed, with the π^* peak at ≈285 e and the σ^* peak at ≈292 eV. The EELS spectra obtained at the interface exhibited similarities to those of SiC taken from a location far away from the interface. Importantly, the intensities of both the σ^* and π^* peaks decreased, resulting in a concave shape at the interface. The lowest intensity of both the σ^* and π^* peaks was observed at the interface at a position of 17.36 nm, indicating a lower density of carbon atoms compared to the adjacent SiC near the interface. This observation is consistent with the findings from the EDS analysis, thus confirming the agreement between the two techniques. The decrease in intensity of the σ^* peak and the increase in the intensity of π^* peak adjacent to the interface were caused during the Si layer deposition process, likely due to Ar plasma

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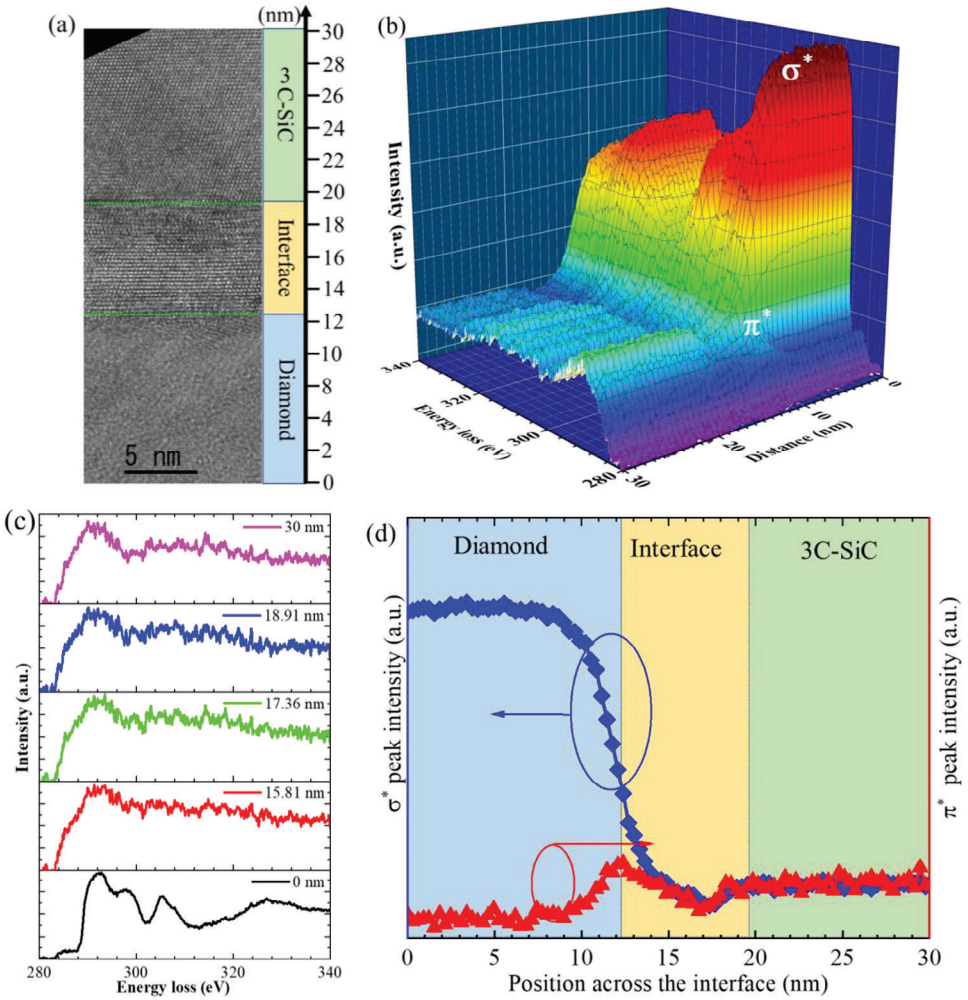


Figure 7. a) A cross-sectional HRTEM image of the 3C-SiC/diamond bonded interface after annealing at 1100 °C, b) EELS line scan across the interface at the C-K edge, c) EELS spectra obtained at a specific distance indicated in (a), d) intensity profiles of the σ*and π*peaks derived from the EELS spectra as a function of the scanned distance.

irradiation. Similar observations were made at the GaN/diamond heterointerface fabricated using the SAB method.[37]

Figure 8a,b shows the picosecond ultrasonics and TDTR measurements on the Al-diamond sample. The echos from the Al-diamond interfaces are used to determine the Al thickness. The used modulation frequency and objective are 9.3 MHz and 5X. The κ of the diamond substrate is measured to be 1964 W m^{-1} K^{-1}, which is close to the thermal conductivity of a perfect single-crystal diamond. The Al-diamond TBC is determined as 150 MW m^{-2} K^{-1}. Figure 8c,d shows the picosecond ultrasonics and TDTR measurements on the Al-3C-SiC-diamond sample. The measured diamond thermal conductivity is used in the data fitting of the Al-3C-SiC-diamond sample. We used a modulation frequency of 1.9 MHz to maximize the sensitivity of the buried TBC. To determine the 3C-SiC thickness, we use the sound velocity of 3C-SiC along [111] direction as 12.5 km s^{-1} from the literature.[49] The 3C-SiC film thickness is measured as ≈958 nm. Figure 8d shows the TDTR data fitting of the experimental ratio signal with the analytical heat transfer solution of the 3C-SiC-diamond sample. The thermal conductivity of the 3C-SiC thin film is measured to be 193 W m^{-1} K^{-1}, while the TBC of the 3C-SiC-diamond interface is determined to be 55 MW m^{-2}·K^{-1}. This TBC value is high enough for efficient device cooling.[50]

The electrical properties of GaN HEMTs on the diamond substrate were characterized and compared with those on the Si and SiC substrates. The sheet resistance (R_{sh}), contact resistance (R_c), and concentration (N_s) of GaN HEMTs on the diamond substrate were measured to be 330 ± 20Ω sq^{-1}, 1.2 ± 0.2 Ω mm, and 8.0 ± 0.1 × 10^{12} cm^{-2}, respectively. Similarly, the values for R_{sh}, R_c, N_s of GaN HEMTs on the Si and SiC substrates were found to be 340 ± 20 and 350 ± 30 Ω sq^{-1}, 1.3 ± 0.1 and 1.4 ± 0.1 Ωmm, and 7.9 ± 0.1 × 10^{12} and 7.8 ± 0.1 × 10^{12} cm^{-2}, respectively. These results are summarized in **Table** 1. Notably, no significant differences were observed in these values among the three substrates, indicating that the transfer process of AlGaN/GaN/3C-SiC layers did not affect the electrical properties of GaN HEMTs.

The drain current (I_D)–drain bias voltage (V_{DS})–temperature characteristics of GaN HEMTs on the Si, SiC, and diamond substrates are shown in **Figure** 9a–c, respectively, while the

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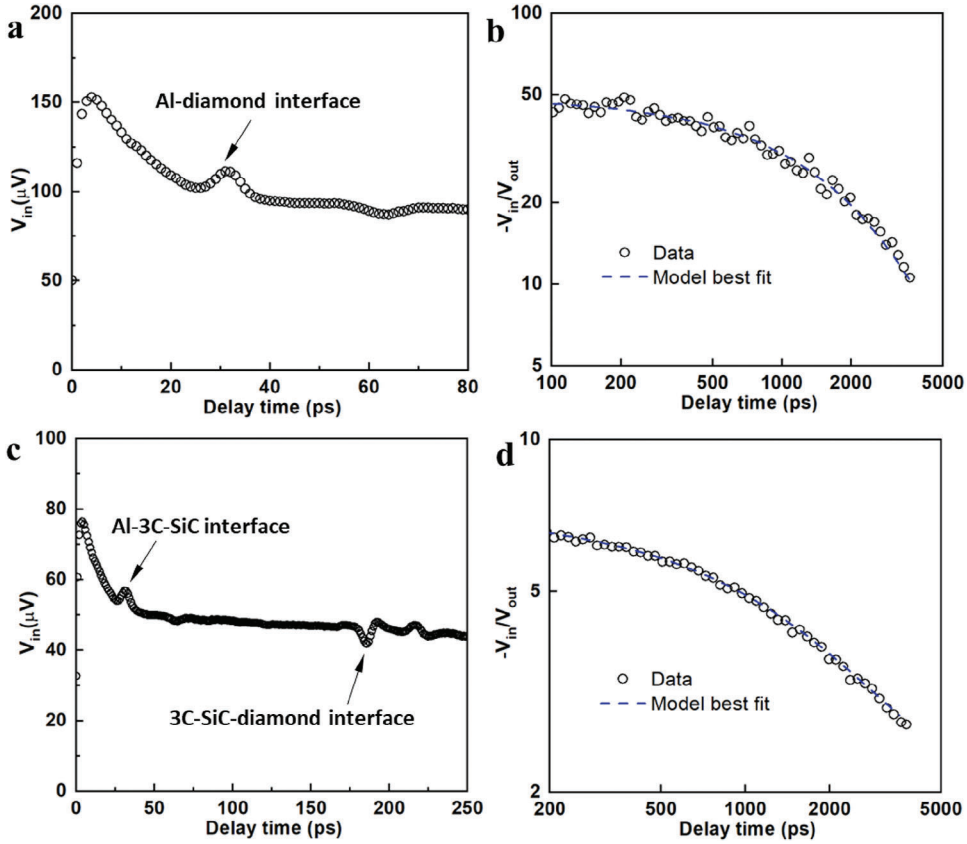


Figure 8. TDTR measurements on the 3C-SiC-diamond sample. a) Picosecond ultrasonics measurements on the diamond substrate sample, b) the data fitting of TDTR experimental data on the diamond substrate sample, c) picosecond ultrasonics measurement on the 3C-SiC-diamond sample, and d) the data fitting of TDTR experimental data on the 3C-SiC-diamond sample.

I_D–V_{DS}–transconductance characteristics of GaN HEMTs on the Si, SiC, and diamond substrates are shown in Figure 9d. The GaN HEMTs have a gate length (L_G) of 5 μm, a gate width (W_G) of 300 μm, a source–gate spacing (L_{SG}) of 10 μm, and agate–drain spacing (L_{GD}) of 10 μm. The gate bias voltage (V_{GS}) was varied between −5 and 2 V with a 0.5 V step. In the I_D–V_{DS} characteristics, excellent pinch-off characteristics were obtained for GaN HEMTs on the Si, SiC, and diamond substrates. In the I_D–V_{DS} curve of GaN HEMTs on the Si and SiC, a peak I_D of 395 and 451 mA mm^{-1}, respectively, was observed separately at V_{DS} = 9 V. For higher V_{DS} regions, I_D decreased with increasing V_{DS}. In contrast, for GaN HEMTs on the diamond substrate, a higher peak of 478 mA mm^{-1} was observed at the same V_{DS} of 9 V, which is higher than those of the GaN HEMTs on the Si and SiC substrates. Furthermore, for the GaN HEMTs on the diamond, even at high V_{GS} and V_{DS} regions, I_D in the saturation region remained almost constant, indicating excellent saturation behavior and stable performance. These results suggest that GaN HEMTs on the diamond substrate exhibit superior current handling capability compared to those on the Si and SiC substrates, with higher peak I_D and more stable saturation behavior even at high bias voltages. This could be attributed to diamonds' high thermal conductivity and the high TBC of the 3C-SiC-diamond interface, making them a promising GaN/3C-SiC-on-diamond structure for high-performance GaN-based devices.

Table 1. Sheet resistance, contact resistance, and concentration of GaN HEMTs.

Materials parameters	GaN HEMTs on Si	GaN HEMTs on SiC	GaN HEMTs on diamond
R_{sh} [Ω sq^{-1}]	340 ± 20	350 ± 30	330 ± 20
R_c [Ωmm]	1.3 ± 0.1	1.4 ± 0.1	1.2 ± 0.2
N_s [10^{12} cm^{-2}]	7.9 ± 0.1	7.8 ± 0.1	8.0 ± 0.1

The temperatures of the gate edge in the gap between the gate and drain at V_{GS} = 2 V and V_{DS} = 20 V for GaN HEMTs on the Si, SiC, and diamond are measured to be 169, 130, and 74 °C, respectively. The temperature of GaN HEMTs on the diamond is 57% lower than those of the GaN HEMTs on the Si and SiC. The threshold voltages of GaN HEMTs on the Si, SiC, and diamond are measured to be −4.2, −4.4, and −4.3 V, respectively, with a small difference that is within the uncertainty of the measurements. The maximum transconductance (g_{m_max}) of GaN HEMTs on the diamond is 76 mS mm^{-1} at V_{GS} = −0.3 V, which is 1.1 times larger than that (g_{m_max} = 69 mS mm^{-1} at V_{GS} = −1.4 V) of GaN HEMTs on the SiC and 1.46 times larger than that (g_{m_max} = 52 mS mm^{-1} at V_{GS} = −2.2 V) of GaN HEMTs on the Si.

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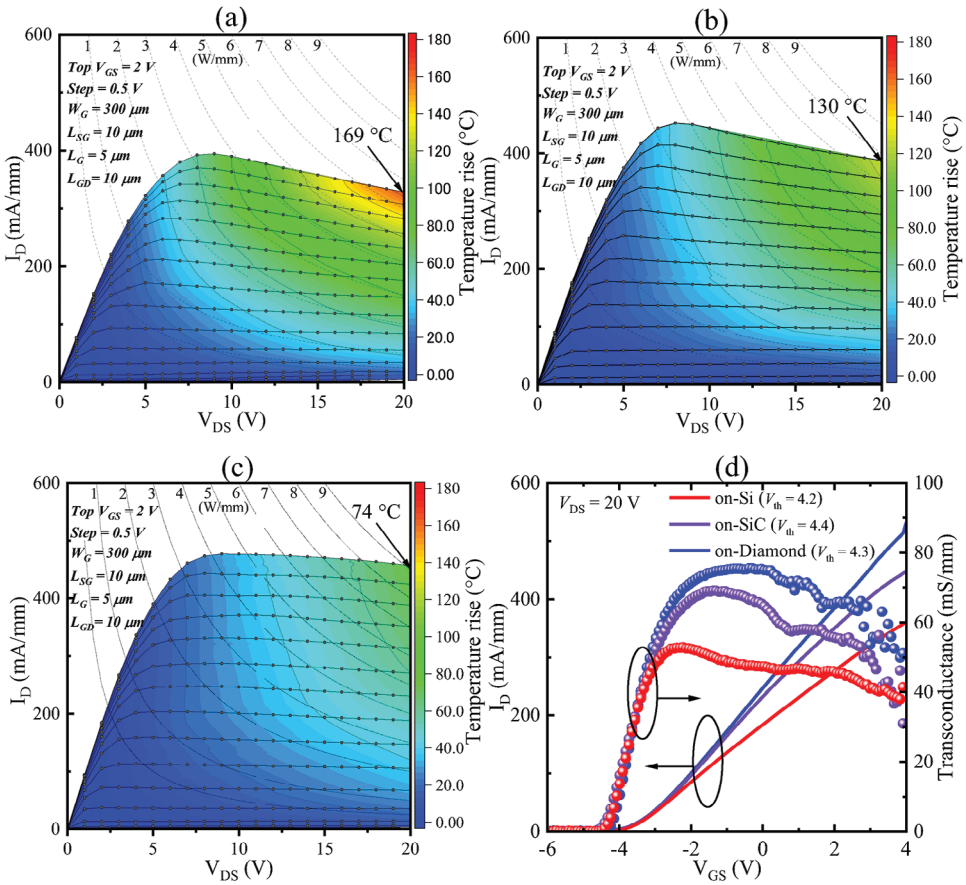


Figure 9. I_D-V_{DS} characteristics of the GaN HEMTs on the a) Si, b) SiC, and c) diamond with L_G = 5 μm, L_{SG} = L_{GD} = 10 μm, and W_G = 300 μm for V_{GS} varied between −5 and 2 V with a step of 0.5 V; d) I_D-V_{GS}-transconductance characteristics of the GaN HEMTs on the Si, SiC, and diamond at V_{DS} = 20 V.

The temperature distribution in the gap between the source and gate, as well as the gap and drain in GaN HEMTs on the Si, SiC, and diamond under different power dissipation (P_{diss}) values (2, 4, 6, and 8 W mm^{-1}), are shown in **Figure 10**a–e. The results show that the temperature rises significantly at the gate edge in the gap between the gate and drain due to the high electric field applied. The highest temperature is observed at the gate edge of GaN HEMTs on the Si, while the lowest temperature is observed at the gate edge of GaN HEMTs on the diamond. Before applying power, the temperatures of the gap between the gate and drain in GaN HEMTs on the Si, SiC, and diamond were measured to be ≈27 °C. When a P_{diss} of 8 W mm^{-1} was applied, the temperature of GaN HEMTs on the diamond increased to 81°C (temperature rise ΔT = 52°C), whereas it increased to 259 and 151.5°C (temperature rise ΔT = 232°C and ΔT = 125°C) on the Si and SiC, respectively. The temperature rise at the gate edge of GaN HEMTs is proportional to P_{diss}. By examining the slopes in Figure 10e, the thermal resistance (R_{TH}) of GaN HEMTs on the Si, SiC, and diamond can be determined. The extracted values are 38, 23, and 10 K mm W^{-1}, respectively. It is noteworthy that the R_{TH} of GaN HEMTs on the diamond is <1/2 of that on the SiC and ≈1/4 of that on the Si substrate, indicating the superior thermal performance of GaN HEMTs on the diamond. The thermal resistance reduction rate of GaN HEMTs on diamond compared to the SiC substrates, as well as the thermal resistance reduction rate of other GaN HEMTs on diamond structures compared to SiC is shown in Figure 10. Notably, we found that the thermal resistance reduction rate in this work is significantly higher than that of other GaN HEMTs on diamond compared to SiC.

A significant difference in thickness was observed for the intermediate layer formed at the interface before and after annealing. This difference can be attributed to two main factors. First, the initially amorphous SiC layer underwent recrystallization because of the Ar fast atom beam irradiation. This process caused the amorphous SiC to transform into a crystalline structure. The phenomenon of recrystallization in the amorphous layer induced by plasma irradiation has been demonstrated in previous studies,[19,45,53] and it is a well-known phenomenon. Second, after annealing, a new SiC layer formed through the reaction between Si and carbon atoms. Previous studies have reported the transformation of the amorphous carbon and amorphous Si layers at the interface fabricated using the SAB method into a SiC layer after a high-temperature annealing process.[54–56] The carbon atoms involved in this reaction originated from the amorphous carbon layer formed on the diamond surface due to Ar plasma irradiation. The formation of the amorphous carbon layer on the diamond surface following plasma irradiation has been reported in the literature.[57,58–60] It is worth noting that the atom density of the amorphous layer is typically smaller than that of the crystal layer. Both processes, the recrystallization of SiC and the reaction

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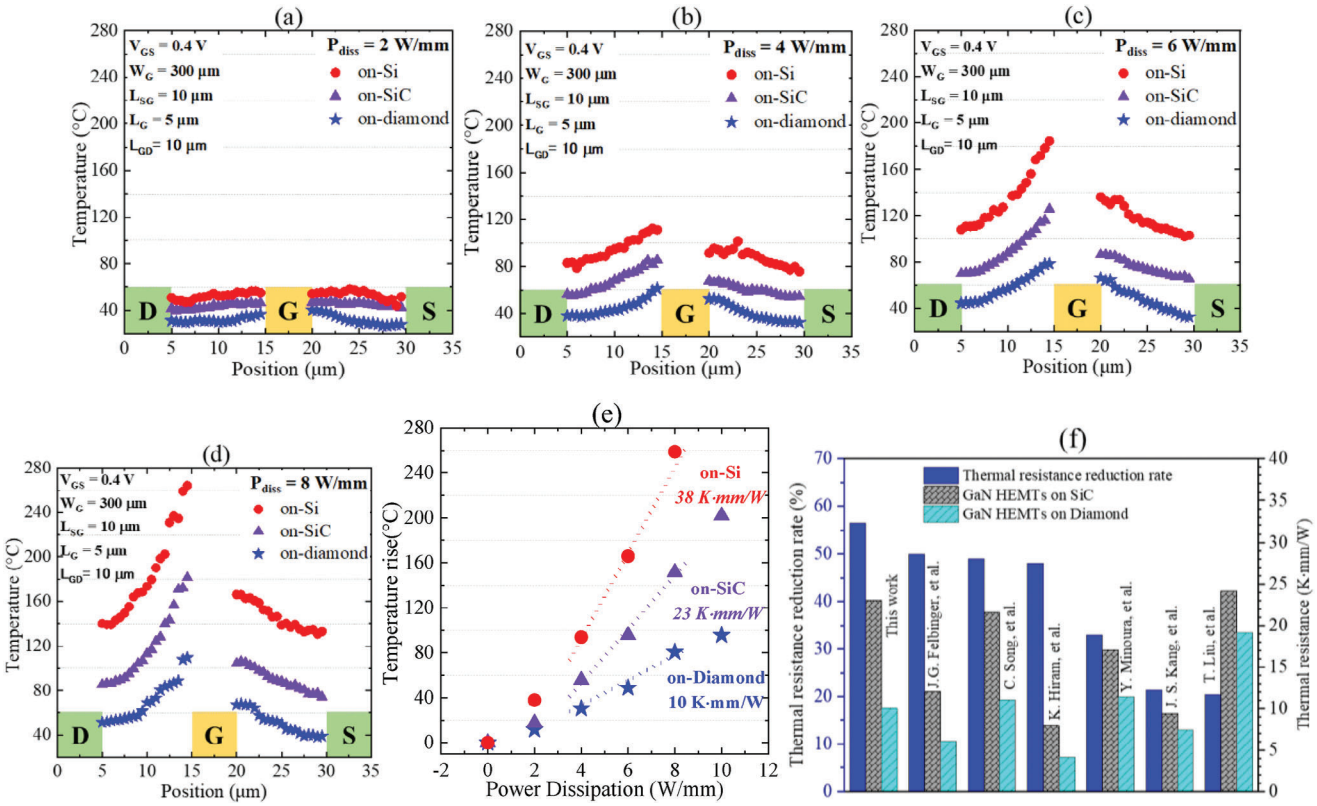


Figure 10. The temperature distribution of the GaN HEMTs on the Si, SiC, and diamond at P_{diss} of a) 2, b) 4, c) 6, and d) 8 W mm^{-1}; e) relationships between the temperature rise of the gate edge and P_{diss} for the GaN HEMTs on the Si, SiC, and diamond; f) the thermal resistance reduction rate of GaN HEMTs on diamond compared to the SiC substrates, as well as the thermal resistance reduction rate of other GaN HEMTs on diamond structures compared to SiC.[6,11,12,24,51,52]

between Si and carbon atoms, contributed to the transformation of the intermediate layer, resulting in the observed discrepancy in thickness.

The lattice fringes observed and the FFT analysis confirm that the SiC layer formed at the interface is indeed crystalline and specifically of the 3C-SiC crystal structure. However, the observed lower densities of carbon and silicon, as well as the wider spacing of the {111} planes parallel to the bonding interface and the narrower spacing of the {111} planes perpendicular to the interface, indicate the presence of strain within the SiC layer formed at the interface. Strain in a crystalline material refers to the distortion or deformation of the crystal lattice from its ideal structure. In this case, the observed strain in the SiC layer suggests the existence of internal stresses within the material. These internal stresses could result from the lattice mismatch between the 3C-SiC layer and the diamond substrate or differences in thermal expansion coefficients. It is worth noting that despite the presence of strain, no exfoliation or peeling of the bonded layers was observed, indicating a strong interface between the AlGaN/GaN/3C-SiC layers and the diamond substrate. Additionally, the observation that the tensile stress of the 3C-SiC layer decreases with increasing annealing temperature suggests that the tensile stresses within the 3C-SiC layer are partially relieved or relaxed during the annealing process. The presence of internal stresses within the SiC layer formed at the interface can have important implications for the overall structural integrity and performance of the bonded layers. The neutralization of tensile stress in the 3C-SiC and GaN layers by the internal stresses of the SiC layer formed at the interface suggests a complex stress distribution within the material system. Further analysis and investigation are indeed necessary to fully understand the formation mechanisms of the SiC layer at the interface. This will provide a comprehensive understanding of the formation mechanisms of the SiC layer and contribute to the development of optimized bonding techniques and improved material properties. The works are already being conducted and the results of the investigations will be reported.

The SiC layer formed at the interface served as a stress-relieving buffer between the materials with different lattice parameters and thermal expansion coefficients, such as 3C-SiC, GaN, and diamond. This buffer layer played a crucial role in accommodating the strain and reducing the overall stress at the bonding interface. By incorporating the 3C-SiC layer, it becomes possible to fabricate large-area bonding and fully exploit the excellent thermal conductivity of diamond substrates. The SiC layer acted as a bridge, allowing for efficient heat dissipation and thermal management in GaN-based devices. Moreover, the presence of the SiC layer contributes to high-quality crystal growth by providing a compatible lattice structure for the epitaxial growth of GaN. This helps to minimize the formation of defects and improve the overall device performance and reliability. Overall, the study presented a promising solution for the mass production of GaN-based devices on diamond substrates. Utilizing the SiC

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layer at the bonding interface, offered effective thermal management, enhanced heat dissipation, and improved crystal growth while minimizing the detrimental effects of thermal stress and defects associated with the bonding process. Further research and development in this area could pave the way for the commercialization of GaN-based devices on diamond substrates for various applications.

4. Conclusion

We have successfully achieved the fabrication of GaN HEMTs on diamond substrates using the bonding first approach. The process involved transferring the AlGaN/GaN/3C-SiC layers grown on Si to the diamond substrate, followed by the fabrication of GaN HEMTs. Before annealing, an amorphous layer with a thickness of ≈15.5 nm was observed at the interface between the 3C-SiC and diamond. However, after annealing at 1100 °C, the amorphous layer underwent recrystallization, and the reaction between carbon and silicon atoms resulted in the formation of a SiC layer. The thickness of the SiC layer formed at the interface was measured to be ≈7.4 nm. Notably, the annealing process also played a significant role in releasing the tensile stress present in the GaN and 3C-SiC layers. This suggests that the high-temperature annealing not only did not lead to interface cracking or defects but also improved the overall quality of the interface. The fabricated GaN HEMTs on the diamond substrate showcased impressive performance characteristics. They exhibited a maximum drain current I_D of 478 mA mm^{-1}, surpassing the performance of GaN HEMTs grown on Si and SiC substrates. The notable enhancement in performance can be ascribed to the high TBC exhibited at the 3C-Sic-diamond interface. This attribute enables the effective utilization of the high thermal conductivity inherent to diamond. Additionally, the GaN HEMTs on the diamond substrate demonstrated significantly reduced thermal resistance compared to those on Si and SiC substrates. This further highlights the superior thermal management capabilities provided by the diamond substrate. The efficient dissipation of heat contributes to the improved performance and reliability of the GaN-based devices. The utilization of GaN/3C-SiC on diamonds holds great promise for the mass production of GaN-based devices on diamond substrates, providing outstanding thermal management capabilities and facilitating the growth of high-quality crystals for superior device performance.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

bonding first, GaN high electron mobility transistors (HEMTs) on the diamond, heat dissipation, high thermal stability, large area diamond bonding, low thermal resistance, thermal management

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